

Module Review: RuvABC Holliday-Junction Processing in *Pseudomonas putida* KT2440

Review topic: `ruvabc_holliday_junction_processing` **Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Anchor pathway:** KEGG ppu03440 — Homologous recombination **Module area:** other_kegg_pathway (no module YAML resolved)

1. Executive Summary

The RuvABC Holliday-junction (HJ) processing module is **complete, syntenically organized, and predicted catalytically competent** in *Pseudomonas putida* KT2440. The three core genes — **ruvA** (PP_1216 / Q88NJ1), **ruvB** (PP_1217 / Q88NJ0) and **ruvC** (PP_1215 / Q88NJ2) — are encoded in a conserved *ruvCAB* arrangement, with *ruvA* and *ruvB* adjacent and *ruvC* directly upstream. RuvC, the resolvase, retains its full RNase-H-fold catalytic active site (annotated acidic residues at positions 8/67/139 and the diagnostic G-I-D-P-G motif), and a global alignment shows **63.4% amino-acid identity to *Escherichia coli* RuvC**, comfortably above the ~30% threshold used for confident ortholog functional transfer. The alternative branch-migration helicase **RecG** (PP_5310 / Q88C73) is also present and intact, providing RuvAB-independent branch migration. On this basis, all core module steps — junction recognition/branch-migration loading (RuvA), ATP-driven branch migration (RuvB, with RecG redundancy), and junction resolution (RuvC) — should be marked **COVERED at homology grade**.

Two caveats are important for curation. First, **all evidence is homology-based**: every core protein carries UniProt PE=3 ("inferred from homology"), and there are no direct KT2440 biochemical or genetic experiments on RuvABC function. Second, the target strain is unusually **DNA-damage-sensitive and recombination-inefficient**: Akkaya et al. (2021) show KT2440 mounts a weak SOS response because of an inefficient RecA–LexA interplay, so the *pathway is present but operates at low efficiency* — a phenotype driven by regulation, not by missing HJ-processing genes.

Finally, the supplied candidate list of 24 genes is **over-broad for this module**. Only *ruvA*, *ruvB*, *ruvC* and (as a related-but-distinct step) *recG* are true HJ-processing genes. DNA polymerase III subunits (*dnaN*, *holC*, *dnaEA*, *holB*, *dnaQ*, *dnaX*, *holA*), *PolA*, *SSB* and the generic "Exonuclease" ORFs belong to replication (*ppu03030*) and should be excluded from the module scope. The presynaptic recombination genes (*recA*, *recBCD*, *recFOR*, *recJ*, *priA*) are legitimate *homologous-recombination* genes but sit **upstream** of HJ processing and belong to sibling module steps, not to `ruvabc_holliday_junction_processing`. We also flag **PP_0151** ("Holliday junction resolvase") as a **spurious machine-generated annotation** that should not be counted as a second resolvase.

2. Target-Organism Pathway Definition

2.1 What the module includes

`ruvabc_holliday_junction_processing` covers the **late steps of homologous recombination and recombinational DNA repair** at which a four-way Holliday junction — formed after *RecA*-mediated strand invasion — is (i) recognized, (ii) branch-migrated to extend heteroduplex DNA, and (iii) resolved into two duplex products by a structure-specific endonuclease. The canonical division of labor, established in *E. coli*, is: **RuvA** recognizes and binds the junction and loads the motor; **RuvB**, a hexameric ATP-driven DNA helicase/pump, drives branch migration; and **RuvC**, an RNase-H-superfamily dimeric endonuclease, resolves the junction by symmetric strand nicking ([PMID: 9501105](#)).

2.2 Neighboring pathways to keep separate

For curation, the module boundary should **exclude**:

- **DNA replication (KEGG *ppu03030* / replisome):** DNA polymerase III holoenzyme subunits (β -clamp *dnaN*, χ *holC*, α *dnaEA*, δ' *holB*, ϵ *dnaQ*, γ/τ *dnaX*, δ *holA*), *Pol I* (*polA*), *SSB* (*ssb*), and generic exonucleases. These are replication-associated and were swept in only because they share the broad "homologous recombination" KEGG map.
- **Presynaptic homologous recombination (upstream of HJ processing):** *RecA* (strand exchange), *RecBCD* and *RecFOR* (end resection / *RecA* loading), *RecJ* (5'→3' ssDNA exonuclease), *PriA* (replication restart). These are *bona fide* HR genes but belong to earlier module steps, not to HJ processing itself.

- **Mismatch repair (ppu03430) and base/nucleotide excision repair (ppu03410/ppu03420):** overlap in some shared nucleases but distinct processes.

2.3 Alternate names / database definitions

- KEGG groups all of these under **ppu03440 "Homologous recombination,"** which is broader than the RuvABC module.
- RuvABC is also described as the **"RuvABC resolvosome"** or **"RuvAB branch-migration complex + RuvC resolvase."**
- RecG is annotated as **"ATP-dependent DNA helicase RecG"** and is sometimes grouped with RuvAB as a parallel branch-migration activity; it is **not** a resolvase.

3. Expected Step Model

Step	Function	Expected gene(s)	KT2440 candidate	Status
1. HJ recognition / motor loading	Binds four-way junction, loads RuvB	<i>ruvA</i>	PP_1216 (Q88NJ1)	Covered
2. Branch migration (motor)	ATP-driven hexameric helicase extends heteroduplex	<i>ruvB</i>	PP_1217 (Q88NJ0)	Covered
3. Junction resolution	Structure-specific endonuclease nicks junction	<i>ruvC</i>	PP_1215 (Q88NJ2)	Covered
2'. Alternative branch migration	RuvAB-independent HJ/ fork branch migration	<i>recG</i>	PP_5310 (Q88C73)	Covered (parallel/ redundant)
3'. Alternative resolvase (RusA-type)	Cryptic/prophage HJ resolvase	<i>rusA</i>	— none	Not expected / absent

Interpretation: The three obligatory steps (1–3) are each covered by a single high-confidence candidate. Step 2 has genuine redundancy via RecG. Step 3 (resolution) depends on **RuvC alone** — there is no second, RusA-type resolvase in the KT2440 candidate set, consistent with RusA being a cryptic prophage function in *E. coli* rather than a conserved core gene.

4. Candidate Genes and Evidence

4.1 Core module genes (promote / keep in module)

ruvA — PP_1216 (Q88NJ1), 205 aa, PE=3. Holliday-junction branch-migration complex subunit RuvA. Encoded immediately adjacent to *ruvB*. Role: junction recognition and RuvB loading. Evidence type: homology/synteny. Caveat: no direct KT2440 assay; PE=3.

ruvB — PP_1217 (Q88NJ0), 348 aa, PE=3. Branch-migration motor. Walker A and Walker B ATP-binding motifs are annotated, consistent with the AAA+/helicase mechanism required for the DNA pump activity. Evidence type: homology + conserved ATPase motifs. Caveat: *E. coli* work (RuvBL268S, PMID: 9973614) shows RuvB can be fully active in vitro yet still fail in vivo, i.e., in-vitro-competent motifs do not guarantee full cellular function — a reason to keep confidence at homology grade.

ruvC — PP_1215 (Q88NJ2), 174 aa, PE=3. Crossover-junction endodeoxyribonuclease (EC 3.1.21.10), the resolvase. Directly upstream of *ruvA/ruvB* in a *ruvCAB* cluster. Retains all three annotated catalytic active-site residues (positions 8, 67, 139) matching the *E. coli* RuvC RNase-H catalytic tetrad (D7/E66/D138/D141); the N-terminal motif M-T-L-I-L-G-I-D-P-G contains the catalytic Asp8. Global Needleman–Wunsch alignment vs. *E. coli* RuvC (P0A814) gives **63.4% identity** (104/164 aligned columns), near-identical length. Evidence type: homology + active-site conservation + high global identity. This is the strongest single line of transfer evidence in the module.

recG — PP_5310 (Q88C73), 692 aa, PE=3. ATP-dependent DNA helicase RecG (EC 5.6.2.4). Both Helicase ATP-binding and Helicase C-terminal domains are intact. Provides RuvAB-independent branch migration and can convert stalled replication forks into Holliday junctions (PMID: 18375550). Note that RecG activity at reversed/stalled forks is itself regulated (e.g., limited by DisA in *Bacillus subtilis*, PMID: 34073022). Curation status: keep as **parallel branch-migration** step, not as a resolvase.

4.2 Upstream HR genes (legitimate, but NOT part of this module)

These are correctly annotated homologous-recombination genes that generate the substrates RuvABC acts on; assign them to their own (presynaptic) module steps rather than to HJ processing:

- **recA — PP_1629 (Q88ME4):** central recombinase / strand exchange.
- **recBCD — recB PP_4673, recC PP_4674, recD PP_4672:** double-strand-break end resection and RecA loading (ExoV).
- **recFOR — recF PP_0012, recO PP_1435, recR PP_4267:** RecA loading at ssDNA gaps.
- **recJ — PP_1477:** 5'→3' ssDNA exonuclease (end processing).
- **priA — PP_5088:** replication restart at recombination intermediates.

All are present and intact, confirming KT2440 encodes a **complete upstream HR machinery** feeding the RuvABC step.

4.3 Replication genes wrongly swept into the candidate set (exclude)

dnaN (PP_0011), *holC* (PP_0979), *dnaEA* (PP_1606), *holB* (PP_1966), *dnaQ* (PP_4141), *dnaX* (PP_4269), *holA* (PP_4796), *polA* (PP_0123), *ssb* (PP_0485), and the generic "Exonuclease" ORFs **PP_0353, PP_4768** are DNA Pol III / replisome / general-repair components (primary buckets ppu03030 / ppu03420). They should be **removed from the HJ-processing module** scope.

4.4 Spurious / over-propagated annotation (do NOT count)

PP_0151 (Q88RH8), 97 aa, PE=4 "Predicted": named "Holliday junction resolvase" **solely** from an automated ProtNLM/Google ML prediction (evidence ECO:0008006), with no curated support. Its domain assignments are **PF09498 (DUF2388, domain of unknown function)**, InterPro IPR012661 / TIGRFAM TIGR02448 ("conserved hypothetical protein"). The sequence begins with a hydrophobic signal-peptide-like stretch (MRYLLPLLFAAAGMASAHAMDT...) and **lacks** the RuvC RNase-H catalytic acidic residues and the diagnostic G-I-D-P-G motif. This is a textbook **annotation-hallucination**: it must not be treated as a second resolvase. Separately, **PP_1116 (Q88NT8)** "Site-specific recombinase, resolvase family" is a serine (Tn3-family) site-specific recombinase, unrelated to HJ resolution in recombinational repair.

5. Gaps, Ambiguities, and Likely Over-Annotations

1. **No experimental evidence in the target strain.** Every core gene is PE=3 (homology-inferred). No KT2440 knockout, complementation, or biochemical resolvase assay exists. This is the dominant knowledge gap; the module is "covered" by homology, not by direct function.
 2. **Single point of failure at resolution.** Resolution depends on RuvC alone — there is **no RusA-type backup resolvase** in the candidate set. This is biologically expected (RusA is a cryptic prophage function in *E. coli*), but it means the module has no redundancy at Step 3. Curators should mark the "alternative resolvase" step as **not_expected_in_target_taxon** rather than a gap.
 3. **Regulatory bottleneck, not a genetic gap.** KT2440 is "very sensitive to DNA damage and displays poor homologous recombination efficiencies," attributed to an inefficient RecA–LexA interplay and weak SOS promoters ([PMID: 33393180](#)). The HJ-processing genes are all present; the phenotype is a *regulation/expression* limitation. This is consistent with earlier reports that *P. putida* SOS/lexA induction differs from *E. coli* ([PMID: 8319897](#); [PMID: 2372559](#)).
 4. **Over-broad candidate list.** 20 of 24 candidates are either replication genes or upstream HR genes. The KEGG ppu03440 "Homologous recombination" bucket is much wider than the RuvABC module and should not be equated with it.
 5. **Over-propagated ML annotation (PP_0151).** As detailed above — exclude.
-

6. Module and GO-Curation Recommendations

Module step	Recommended status	Rationale
RuvA (junction recognition/loading)	covered (homology)	PP_1216, syntenic, PE=3
RuvB (branch-migration motor)	covered (homology)	PP_1217, Walker A/B annotated, PE=3
RuvC (junction resolution)	covered (homology, high confidence)	PP_1215, 63% identity to <i>E. coli</i> , full active site
RecG (alt. branch migration)	covered (parallel/redundant)	PP_5310, domains intact
RusA-type alt. resolvase	not_expected_in_target_taxon	cryptic prophage function; no ortholog
Upstream HR (recA/recBCD/recFOR/recJ/priA)	out of module scope (assign to sibling steps)	present but presynaptic
Pol III / replisome / generic exonucleases	out of module scope (exclude)	belong to ppu03030
PP_0151 "HJ resolvase"	exclude — likely over-annotation	ML-only ECO:0008006, DUF2388, no catalytic residues

Module-boundary judgment: The generic KEGG bucket boundary is **wrong for this module** because it conflates the whole ppu03440 "Homologous recombination" map with the narrow RuvABC step. Recommend authoring a dedicated `ruvabc_holliday_junction_processing` module document scoped to RuvA/RuvB/RuvC (+RecG as a parallel branch-migration alternative), and explicitly listing RusA-type resolution as `not_expected`.

GO terms of interest (no new-term request appears necessary): GO:0009378 (four-way junction helicase / Holliday junction helicase), GO:0008821 (crossover junction endodeoxyribonuclease, RuvC), GO:0000725 (recombinational repair), GO:0006310 (DNA recombination). Existing terms cover the module; the main curation action is **restricting propagation**, not requesting new terms.

7. Genes to Promote to Full fetch-gene Review

Priority order:

1. **ruvC (PP_1215 / Q88NJ2)** — the resolvase and the sole HJ-resolution step; highest curation value. Confirm active-site residues and the 63%-identity ortholog call in a full review.
 2. **ruvB (PP_1217 / Q88NJ0)** — confirm Walker A/B motifs and hexamer-forming residues; the *E. coli* RuvBL268S precedent ([PMID: 9973614](#)) means motif presence ≠ full in-vivo function.
 3. **ruvA (PP_1216 / Q88NJ1)** — confirm junction-binding domains and synteny.
 4. **recG (PP_5310 / Q88C73)** — confirm helicase domains and clarify its "parallel branch migration" (not resolvase) role in the module document.
 5. **PP_0151 (Q88RH8)** — promote specifically to **correct/remove** the spurious "Holliday junction resolvase" name and reclassify as DUF2388 hypothetical protein.
-

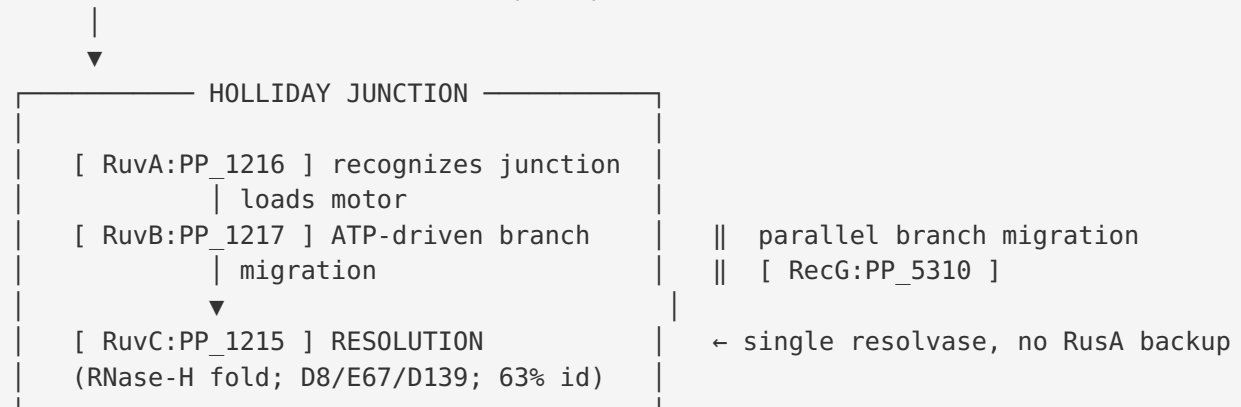
8. Mechanistic Model / Interpretation

Homologous recombination in *P. putida* KT2440

DSB / stalled fork

↓
[RecBCD] or [RecFOR + RecJ] ← end resection / RecA loading (upstream; present, PE≈3)

↓
[RecA] strand invasion → D-loop (present)



↓
Two duplex products → [PriA] replication restart

REGULATORY OVERLAY: weak RecA–LexA/SOS circuit (<https://pubmed.ncbi.nlm.nih.gov/33391111/>)
→ genes all present, but low induction ⇒ poor HR efficiency & DNA-damage sensitivity

The narrative: KT2440 possesses a genetically **complete** HR pathway from resection through resolution. The RuvABC resolvosome is intact and syntenic, RuvC is a high-confidence *E. coli* ortholog, and RecG supplies branch-migration redundancy. The organism's documented fragility toward DNA damage is therefore best explained not by any missing gene, but by a **regulatory bottleneck** in the SOS response that limits how strongly these genes are induced.

9. Evidence Base

PMID	Relevance	How it supports/challenges the review
9501105	RuvABC division of labor (<i>E. coli</i>)	"branch migration is catalysed by the RuvB protein... loaded onto the junction by RuvA, whereas resolution is promoted by the RuvC endonuclease" — defines the roles the KT2440 orthologs perform.
7923356	RuvC atomic structure (<i>E. coli</i>)	"the catalytic center, comprising four acidic residues, lies at the bottom of a cleft" — the acidic residues KT2440 RuvC retains, supporting predicted catalysis.
23118486	<i>T. thermophilus</i> RuvC dimer	Supports the conserved "nick-counter-nick" HJ-resolution mechanism across prokaryotes, applicable to KT2440 RuvC.
27013661	Poxvirus resolvase vs RuvC	Confirms the RNase-H-superfamily active-site/metal-binding architecture RuvC shares.
18375550	RecG as independent branch-migration helicase	"Orthologs of RecG and RuvABC are highly conserved... independent pathways that branch migrate Holliday junctions... RecG... convert stalled replication forks into Holliday junctions."
9973614	RuvBL268S mutant (<i>E. coli</i>)	In-vitro-active RuvB can still fail in vivo — a caution that motif presence ≠ full function; keeps RuvB at homology grade.
33393180	Faulty SOS in KT2440 (target strain)	"very sensitive to DNA damage and displays poor homologous recombination efficiencies"; "fails to mount a strong SOS response due to the inefficacy of the crucial RecA-LexA interplay."
8319897	<i>P. putida/aeruginosa</i> <i>lexA</i> induction	<i>P. putida</i> <i>lexA</i> induction is higher/earlier and UV-dose-independent vs <i>E. coli</i> — corroborates atypical SOS regulation.
2372559	SOS-like response in <i>P. putida</i>	RecA-dependent SOS-like response exists in <i>P. putida</i> , but with distinct behavior.
34073022	DisA limits RecG at forks	RecG fork activity is itself regulated — relevant to interpreting RecG's role.
37358447	RuvABC not required for certain IS processes	Context on RuvABC dispensability in some transposition contexts.

Evidence grade for the target organism: All *P. putida* KT2440 conclusions about RuvABC/RecG gene content and structure are **homology-grade** (UniProt PE=3; sequence/synteny/active-site analysis). The **only direct target-strain experimental evidence** is the SOS/recombination-efficiency phenotype (PMID: 33393180). Functional transfer of RuvABC roles from *E. coli* is rated **strong** for RuvC (63% identity, full active site) and **strong-to-moderate** for RuvA/RuvB/RecG (domain/synteny conservation, but the RuvBL268S precedent tempers certainty).

10. Limitations and Knowledge Gaps

- **No direct KT2440 function data** for any RuvABC/RecG gene (no knockouts, no purified-protein resolvase assays). Confidence rests on homology and conserved motifs.
 - **Single resolvase:** no redundancy at the resolution step; if RuvC were non-functional, the model predicts no backup (RusA-type) resolvase.
 - **Regulation not fully mapped:** the weak SOS circuit is documented at the *recA/lexA* level, but induction kinetics of *ruvCAB* specifically in KT2440 have not been measured.
 - **Sequence analysis performed here** (identity, active-site inspection) is bioinformatic; structural or biochemical confirmation is absent.
-

11. Proposed Follow-up Experiments / Curation Actions

1. **Curation (immediate):** Re-scope the module to {ruvA PP_1216, ruvB PP_1217, ruvC PP_1215} + RecG PP_5310 (parallel branch migration); mark RusA-type resolvase **not_expected_in_target_taxon**; move *recA/recBCD/recFOR/recJ/priA* to sibling presynaptic steps; exclude all Pol III / replisome / generic-exonuclease genes.
2. **Fix annotation:** Submit a correction for **PP_0151** (remove "Holliday junction resolvase"; reclassify as DUF2388 hypothetical) and flag the ProtNLM ECO:0008006 source.
3. **Promote to full review:** *ruvC* → *ruvB* → *ruvA* → *recG* → PP_0151 (as above).
4. **Wet-lab (to resolve the core gap):** construct *ruvC*, *ruvAB*, and *recG* deletions in KT2440; assay UV/mitomycin-C/ciprofloxacin sensitivity and conjugational/recombination frequency to confirm functional roles directly in the target strain.
5. **Biochemistry:** purify KT2440 RuvC and test HJ cleavage in vitro (Mg^{2+}/Mn^{2+} dependence) to validate the RNase-H active-site prediction.

6. **Expression:** measure *ruvCAB* induction after DNA damage in KT2440 to test whether the weak-SOS phenotype extends to reduced RuvABC induction specifically.
-

12. Key References

- Van Gool AJ et al. Functional interactions between the Holliday junction resolvase and the branch migration motor of *E. coli*. [PMID: 9501105](#)
 - Ariyoshi M et al. Atomic structure of the RuvC resolvase. [PMID: 7923356](#)
 - Górna AE et al. Structural asymmetry in the *T. thermophilus* RuvC dimer. [PMID: 23118486](#)
 - Poxvirus resolvase structure/metal binding. [PMID: 27013661](#)
 - Repair and antirepair DNA helicases in *Helicobacter pylori* (RecG/RuvABC). [PMID: 18375550](#)
 - *E. coli* RuvBL268S mutant. [PMID: 9973614](#)
 - Akkaya Ö et al. The faulty SOS response of *P. putida* KT2440. [PMID: 33393180](#)
 - *P. putida/aeruginosa* *lexA* induction. [PMID: 8319897](#)
 - SOS-like reactions of *P. putida*. [PMID: 2372559](#)
 - DisA limits RecG at stalled/reversed forks. [PMID: 34073022](#)
 - RuvABC not required for certain IS processes. [PMID: 37358447](#)
-

Consensus Answer

The RuvABC Holliday-junction processing module is complete and predicted catalytically competent in *P. putida* KT2440: *ruvA* (PP_1216), *ruvB* (PP_1217) and *ruvC* (PP_1215) form a conserved *ruvCAB* cluster; RuvC is a strong *E. coli* ortholog (63% identity, intact RNase-H active site), and RecG (PP_5310) provides parallel branch migration — so all core steps are COVERED at homology grade. All evidence is homology-based (PE=3, no direct KT2440 experiments), and the strain's documented DNA-damage sensitivity reflects a weak RecA–LexA/SOS regulatory circuit rather than any missing HJ-processing gene. The supplied candidate list is over-broad (Pol III/replisome and upstream HR genes belong to other buckets), and PP_0151 "Holliday junction resolvase" is a spurious ML annotation to be excluded.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery